

REQ-ARC-01	CalculateOddParity
REQ-ARC-02	Prototype: bool CalculateOddParity(uint32_t raw) Description: The function shall calculate the parity bit required to provide odd parity for the given input.
REQ-ARC-03	IF All of the following conditions are satisfied: a. raw contains an odd number of set bits Following operation are performed: a. return false
REQ-ARC-04	IF all of the following conditions are satisfied: a. raw contains an even number of set bits Following operation are performed: a. return true
REQ-ARC-05	CheckParity
REQ-ARC-06	Prototype: bool CheckParity(uint32_t raw) Description: The function shall verify whether the given ARINC 429 word satisfies the odd parity requirement.
REQ-ARC-07	IF all of the following conditions are satisfied: a. receivedParity is equal to calculatedParity Following operations are performed: a. return true
REQ-ARC-08	IF all of the following conditions are satisfied: a. receivedParity is NOT equal to calculatedParity Following operations are performed: a. return false
REQ-ARC-09	CreateARINCWord
REQ-ARC-10	Prototype: uint32_t CreateARINCWord(ARINCWord& word) Description: The function shall create a 32-bit ARINC 429 word from the provided ARINCWord structure.
REQ-ARC-11	IF all of the following conditions are satisfied: a. word is a 32 bit ARINC 429 word. Following operations are performed: a. place the Label field in Bits 0-7 b. place the SDI field in Bits 8-9 c. place the Data field in Bits 10-28 d. place the SSM field in Bits 29-30 e. calculate odd parity bit using bits 0-30 and place the resulting parity bit in Bit 31. f. return 32 bit ARINC 429 word.
REQ-ARC-12	DecodeARINCWord
REQ-ARC-13	Prototype: ARINCWord DecodeARINCWord(uint32_t rawWord)

REQ-ARC-13	Description: The function shall decode a 32-bit ARINC 429 word into an ARINCWord structure.
REQ-ARC-14	IF all of the following conditions are satisfied: a. rawWord is a 32-bit ARINC 429 word Following operations are performed: a. extract Bits 0–7 from rawWord and assign the value to the label field b. extract Bits 8–9 from rawWord and assign the value to the sdi field c. extract Bits 10–28 from rawWord and assign the value to the data field d. if Bit 28 is set, sign extend the data field e. extract Bits 29–30 from rawWord and assign the value to the ssm field f. extract Bit 31 from rawWord and assign the value to the parity field g. return the decoded ARINCWord structure
REQ-ARC-15	GenerateSSM
REQ-ARC-16	Prototype: SSM GenerateSSM() Description: The function shall generate a random integer between 1 and 100 and assign an SSM value according to the generated number.
REQ-ARC-17	IF all of the following conditions are satisfied: a. roll is equal to 1 Following operations are performed: a. return SSM::FailureWarning
REQ-ARC-18	IF all of the following conditions are satisfied: a. roll is equal to 2 Following operations are performed: a. return SSM::NoComputedData
REQ-ARC-19	IF all of the following conditions are satisfied: a. roll is equal to any other value between 3 and 100 Following operations are performed: a. return SSM::NormalOperation